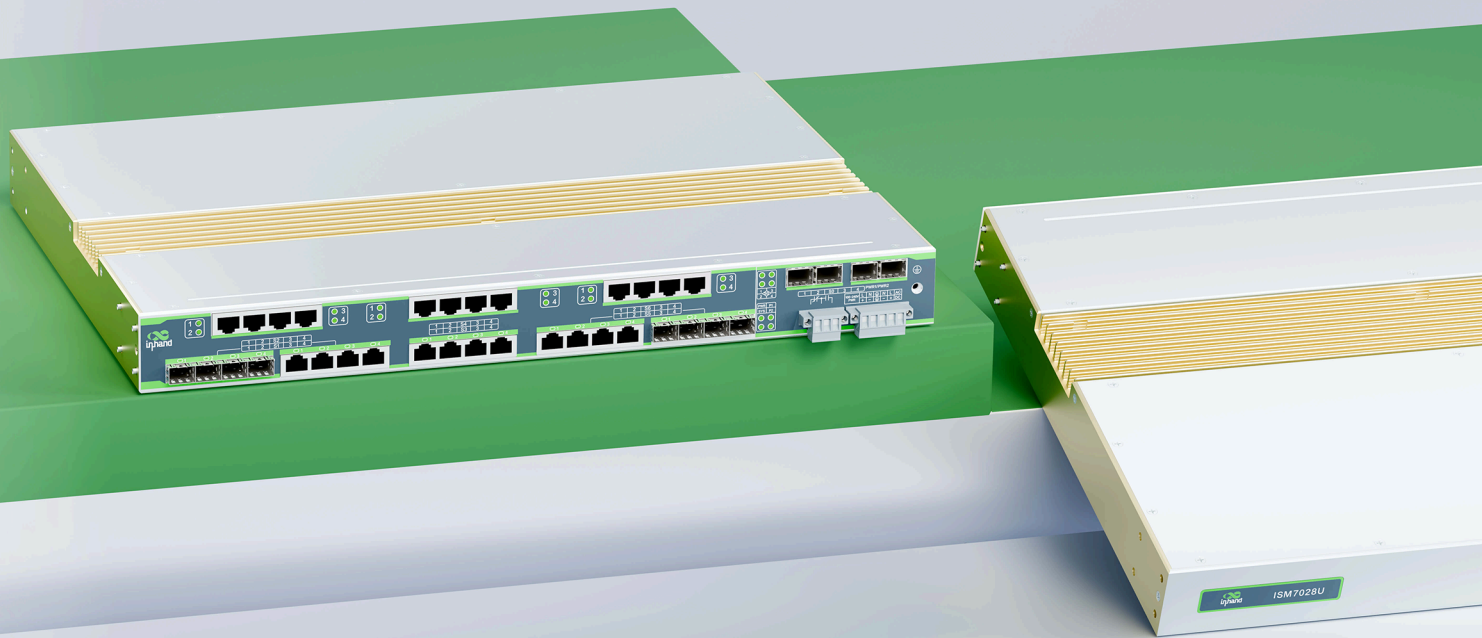


Build advanced and highly reliable industrial Ethernet communication system



ISM7028U-P Managed Industrial Ethernet Switch

· Public Utility

· Smart Manufacturing

· Smart City

· Smart Energy

1. Product Overview

The ISM series of managed industrial Ethernet switches is specifically designed to meet the demands of harsh industrial environments such as power, transportation, and industrial control applications. The rugged casing design and protected circuits among other industrial-grade features perform its remarkable adaption to rough environments.

Product Features:

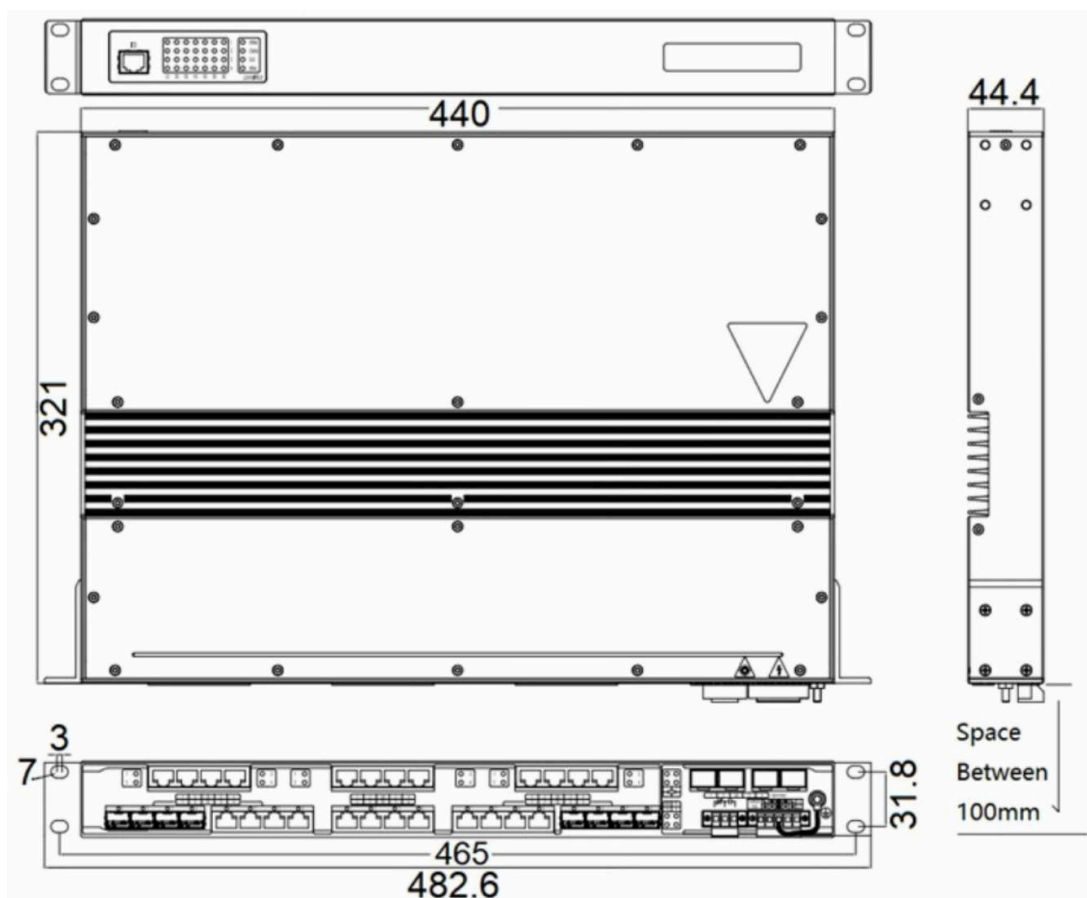
- **Reliability & Resilience:** Fanless design, IP40 protection rating, robust metal casing, dust and dirt resistance, and wide temperature operation (-40 °C ~ +85 °C (-40 °F ~ +185 °F)). Industrial-grade dual redundant high-voltage power supply design.
- **Redundancy Protocols:** Supports STP/RSTP/MSTP ring redundancy protocols offering flexible choices for constructing intricate communication systems. Any two ports can establish an independent self-recovery ring.
- **Network Management:** Supports SNMP for integrated network management and RMON for effective monitoring. Allows unified and quick deployment in rack mount formats.
- **Enhanced Security & Control:** Comprehensive linear industrial switch functions, offering high performance, data packet dropout protection, and intelligent features such as VLAN network segmentation and advanced multicast protection mechanisms.

Key Technical Specifications

Parameter	Specification
Type	Layer 3 Managed Industrial Ethernet Switch
Ports	16x 10/100/1000BaseT + 8x 100/1000Base(X/T) COMBO + 4x 1000/2500BaseX SFP
Switching Performance	128 Gbps backplane bandwidth; 16K MAC table; 12 Mbit buffer

Parameter	Specification
Dimensions / Weight	440 × 321 × 44.4 mm (17.32 × 12.64 × 1.75 in) / 4.1 kg (9.04 lb)
Power	100-240V AC dual-isolated inputs
Environment	-40 °C ~ +85 °C (-40 °F ~ +185 °F) operating; IP40; fanless
Management	Web, CLI, SNMPv1/v2c/v3, RMON
Protocols	STP/RSTP/MSTP, IGMP Snooping, OSPF, VRRP, PIM-SM/DM, BGP, Port Trunking
EMC / Certifications	FCC Part 15 Class A; EN55022 Class A; IEC 61000-4 series; IEC61850-3

2. Product Dimensions



ISM7028U

Note:

1. All dimensions are in millimeters (mm), with inches (in) in parentheses.
2. All dimensions are approximate and **for reference only**.
3. The dimensions shown in the figure **shall not be used for production or processing**.
4. Dimensions must comply with part and manufacturing tolerance requirements.
5. Dimensions are subject to change without notice.

3. Technical Specifications

3.1 Protocol Compliance List

Category/Parameter	Specification
IEEE Standards	
IEEE 802.3	CSMA/CD method and physical Layer specifications
IEEE 802.1p	Priority Queuing
IEEE 802.1q	VLAN tagging
IEEE 802.1d	Spanning Tree Algorithm
IEEE 802.1w	Rapid Spanning Tree
IEEE 802.1s	Multiple Spanning Tree
IEEE 802.3ac	VLAN Tagging
IEEE 802.1x	Authentication
IEEE 802.3ad	Link Aggregation
IEEE 802.3x	Flow Control
IEEE 802.3	Ethernet
IEEE 802.3u	Fast Ethernet

Category/Parameter	Specification
IEEE 802.3z	Gigabit Ethernet
IEEE 802	Networks
RFC Standards	
RFC 213	DHCP Server
RFC 768	UDP
RFC 791	IP
RFC 792	ICMP
RFC 793	TCP
RFC 826	ARP
RFC 854	Telnet Client & Server
RFC 904	Exterior Gateway Protocol Formal Specification
RFC 1027	Using ARP to Implement Transparent Subnet Gateways
RFC 1058	RIP
RFC 1059, 1119	NTPv1/2
RFC 1112	IGMP
RFC 1191	Path MTU Discovery
RFC 1256	ICMP Router discovery protocol
RFC 1267	A Border Gateway Protocol 3 (BGP-3)
RFC 1388	RIP Version 2 Carrying Additional Information
RFC 1403	BGP OSPF Interaction
RFC 1519	CIDR (Classless Inter-domain Routing)

Category/Parameter	Specification
RFC 1587	OSPF NSSA
RFC 1765	OSPF Database Overflow
RFC 1812	Requirements for IP Version 4 Routers
RFC 1994	PPP Challenge Handshake Authentication Protocol (CHAP)
RFC 2068	HTTP
RFC 2138	RADIUS
RFC 2139	RADIUS Accounting
RFC 2236	IGMPv2
RFC 2328	OSPF V2
RFC 2338	VRRP
RFC 2362	PIM-SM/DM
RFC 2370	The OSPF Opaque LSA Option
RFC 2474	DiffServ Precedence
RFC 2475	DiffServ Core and Edge Router Functions
RFC 2597	DiffServ Assured Forwarding
RFC 2598	DiffServ Expedited Forwarding
RFC 2644	Directed Broadcasts
RFC 2865	Remote Authentication Dial In User Service (RADIUS)
RFC 3046	DHCP Relay Agent Information Option
RFC 3222	Forwarding Information Base (FIB)
Other Protocols	

Category/Parameter	Specification
GMRP	GARP
GVRP	GARP
SSH2	Secure Shell 2
IGMP	Snooping
SNMPv3	Supported

3.2 Hardware Specifications

Category/Parameter	Specification
Physical Performance	
Enclosure	Fully enclosed seamless metal enclosure
Dimensions (W × D × H)	440 mm × 321 mm × 44.4 mm
Weight	4.1 kg (9.04 lb)
Mounting Method	Rack mounting
Cooling Method	Fanless cooling
Ingress Protection	IP40
Storage Temperature	-40 °C ~ +85 °C (-40 °F ~ +185 °F)
Operating Temperature	-40 °C ~ +85 °C (-40 °F ~ +185 °F)
Humidity	5 ~ 95% (non-condensing)
Hardware Performance	
Backplane Bandwidth	128 Gbps

Category/Parameter	Specification
Transmission Mode	Parallel Storage Forwarding
MAC Table Size	16K
Packet Buffer Size	12 Mbits
Exchange Rate	148,800 pps/100M ports; 1,488,000 pps/1000M ports
Power Parameters	
Input Voltage	100-240V dual-isolated power inputs
Electromagnetic Characteristics	
EMI	FCC 47 CFR Part 15 Class A; EN55022 Class A
EMS	IEC(EN)61000-4-2, Class 4 IEC(EN)61000-4-3, Class 3 IEC(EN)61000-4-4, Class 4 IEC(EN)61000-4-5, Class 4 IEC(EN)61000-4-6, Class 3 IEC(EN)61000-4-9, Class 5
Mechanical Characteristics	
Shock	IEC60068-2-27
Freefall	IEC60068-2-31
Vibration	IEC60068-2-6
Certifications	
Certifications	IEC61850-3
Quality Assurance	
Warranty Period	5 years

Category/Parameter	Specification
MTBF	35 years

4. Software Specifications

Software Functions	Specification
Redundancy	STP, MSTP, RSTP, Port Trunking
Management Mode	Browser, Serial Port, STD-17 MIB-II, STD-58 SMIV2, STD-59 RMON, STD-62 SNMPv3, SNMPv2c, SNMPv1
Time Synchronization	NTP
Diagnostic Mode	Indicator light, Journal File, RMON, Port Mirroring, TRAP
Others	IPv4/IPv6 multicast, storm control, MC/BC protection, support Jumbo Frame

5. Ordering Guide

Model	Description
ISM7028U-P-8GC-4GSFP-16GT-HV-HV	28-port Layer 3 managed Industrial Switch. 16 <i>10/100/1000BaseT Ports</i> , 8 <i>100/1000Base(X/T) COMBO Ports</i> , and 4 * <i>1000/2500BaseX SFP Ports</i> (SFP module not included), dual isolated 100-240V power inputs.

6. Contact Us

- Website: [InHand Networks](#)
- Copyright: ©InHand Networks All rights reserved